



JAVA PROGRAMMING

(340)

STATE 2022

Production:

Tollway Customer Database State _____ (425 points)

TOTAL POINTS _____ **(425 points)**

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

You will have ninety (90) minutes to complete your work.

Your name and/or school name should *not* appear on work you submit for grading.

1. Create a folder on the flash drive provided using your contestant number as the name of the folder.
2. Copy your entire solution/project into this folder.
3. Submit your entire solution/project so that the graders may open your project to review the source code.
4. Ensure that the files required to run your program are present and will execute on the flash drive provided.

*Note that the flash drive letter may *not* be the same when the program is graded as it was when you created the program.

*It is recommended that you use relative paths rather than absolute paths to ensure that the program will run regardless of the flash drive letter.

The graders will *not* compile or alter your source code to correct for this.
Submissions that do *not* contain source code will *not* be graded.

Assumptions to make when taking this assessment:

- The input file will contain only text and use “,” as a delimiter.
- A test input file will be available and named, "Names.txt." This file needs to be in your project folder and will be used to help with the solution. This file will be accessed by the TollwayCustomerDataBaseState class which will be the file that you create.

Development Standards:

- Your Code must use a consistent variable naming convention.
- All subroutines (if any), functions (if any), and methods (if any) must be documented with comments explaining the purpose of the method, the input parameters (if any), and the output (if any).
- If you create a class, then you must use Javadoc comments.

Note to Graders: The contestants have been given a file named Names.txt. You must first delete that file from the contestant's flash drive and run the program for Test case #1. This should result in a message that the file was not found. Then replace the file on their flash drive with the grader's file with the same name and run Test case #2. You'll run their program with a grader's file named Names.txt which will contain different data than the student file. This will produce different results than the student's file. You must copy the grader's Names.txt file into the folder where their program runs. Listed below are the test cases and their results.

Test case #1 verifies that the program checks for the correct input file. The output wording does *not* need to be exact.

Text case #2 processes the correct data file name and produces output. The output wording *must* be exact.

Test Case #1

The file doesn't exist.

Test Case #2

>>>>>>>>>>> There were 12 created.<<<<<<<<<<<<<

Name: Jose Montana
Car Information: Ford F150 ABC-123
Payment: \$65.00 WHJFHJFO483897

Name: Isaac Newton
Car Information: Chevy Silverado DFR-345
Payment: \$89.01 WH24223442897

Name: Albert Einstein
Car Information: Honda Accord DWP-098
Payment: \$11.00 454FGGF2897

Name: Carl Gauss
Car Information: Toyota Prius QQQ-123
Payment: \$456.55 JGHJF23442897

Name: Alan Turing
Car Information: Volvo XC90 HJH-345
Payment: \$45.33 GHGDHD3442897

Name: Rene Descartes
Car Information: BMW X1 RWW-345
Payment: \$99.01 EQRET23442897

Name: Ada Lovelace
Car Information: Mercedes G-Class 567-WWW

Payment: \$345.67 DFGDFDFF3442897

Name: Sofya Kovalevskaya

Car Information: Lexus GX350 W4R-R5R

Payment: \$34.56 454FDGHSFDKH897

Name: Benjamin Banneker

Car Information: Lexus RX350 PPR-15R

Payment: \$62.46 WEWE2344GSFRG

Name: Fern Hunt

Car Information: Honda Pilot FGW-011

Payment: \$31.00 MGFNH2344TETR7

Name: Mark Dean

Car Information: Toyota 4Runner WWW-123

Payment: \$152.55 222EREG2897

Name: Katherine Johnson

Car Information: Mercedes GLS-Class 756-QQQ

Payment: \$3,445.67 5HBH5B23442897

Your application will be graded on the following criteria:		
Solution and Project		
The project is present on the flash drive	_____	10 points
The projects main class is named TollwayCustomerDataBaseState	_____	10 points
The projects helper classes are named Customer, Name, CarInfo, Payment (5 Points each)	_____	20 points
Program Execution		
The program runs from the USB flash drive	_____	15 points
If the program does <i>not</i> execute, then the remaining items in this section receive a score of zero.		
The TollwayCustomerDataBaseState class runs and reads the input file	_____	30 points
The TollwayCustomerDataBaseState class displays an error message if the file cannot be found	_____	20 points
The TollwayCustomerDataBaseState class stores data fields from Names.txt into proper variables	_____	20 points
The StockDriver class stops reading the input file when there is no more text	_____	20 points
Program displays the correct output in the correct order for the records and the field with the formatting from the helper classes.	_____	30 points
The output should count and display 12 created records	_____	20 points
The deposit value was formatted for US currency including “\$” and “,” as well as the cents.	_____	10 Points
Source Code Review		
The source code is properly commented		
A comment containing the contestant number is present	_____	10 points
Methods and code sections are commented	_____	20 points
The TollwayCustomerDataBaseState class creates Customer objects from data contained in Names.txt	_____	30 points
Each Customer constructor is comprised of parameters from instantiated objects: Name, CarInfo, and Payment	_____	30 points
Name, CarInfo, and Payment objects correctly receive their information from the Names.txt file	_____	20 points
BufferedReader getReader(String name): Code uses try... catch for exception handling in TollwayCustomerDataBaseState class when reading the file	_____	20 points

Total Points = ____ / 425 points

[illegible]

```

31
32 //Reads the information from the file and breaks it into each of the objects that go into the Customer object
33 private static Customer readCustomer(BufferedReader in)
34 {
35     //Name Class plus Fields
36     Name name; String firstN; String lastN;
37     //CarInfo Class plus Fields
38     CarInfo carinfo; String make; String model; String plates;
39     //Payment Class plus Fields
40     Payment payment; double deposit; String creditcard;
41     String line = "";
42     String[] data;
43
44     //catches errors in the read
45     try{
46     {
47         line = in.readLine();
48     }
49     catch (IOException e)
50     {
51         System.out.println("I/O Error");
52         System.exit(0);
53     }
54     //ends the read once it gets to the end
55     if (line == null)
56         return null;
57     else
58     {
59         data = line.split(","); //"," is the delimiter to separate the fields for the records
60         firstN = data[0]; //each field goes in escalating arithmetic sequence 0 to 6
61         lastN = data[1];
62         make = data[2];
63         model = data[3];
64         plates = data[4];
65         deposit = Double.parseDouble(data[5]);
66         creditcard = data[6];
67         //construct the objects to be placed in the Customer object
68         name = new Name(firstN, lastN);
69         carinfo = new CarInfo(make, model, plates);
70         payment = new Payment(deposit, creditcard);
71         return new Customer(name, carinfo, payment);
72     }
73 }
74
75 //imports the file properly for BufferedReader
76 private static BufferedReader getReader(String name)
77 {
78     BufferedReader in = null;
79     try
80     {
81         File file = new File(name);
82         in = new BufferedReader(
83             new FileReader(file) );
84     }
85     catch (FileNotFoundException e)
86     {
87         System.out.println(
88             "The file doesn't exist.");
89         System.exit(0);
90     }
91     return in;
92 }
93 }
94
95

```

```

95
96 //State Customer Information
97 class Customer
98 {
99     public Name name;
100     public CarInfo carinfo;
101     public Payment pay;
102
103     public Customer(Name n, CarInfo c, Payment p)
104     {
105         this.name = n;
106         this.carinfo = c;
107         this.pay = p;
108     }
109
110     public String getInfo(){
111
112         return this.name.getInfo() + "\n" + this.carinfo.getInfo() + "\n" + this.pay.getInfo() + "\n" ;
113     }
114 }
115
116 class Name
117 {
118     private String last_Name;
119     private String first_Name;
120
121     public Name(String fn, String ln)
122     {
123         this.last_Name = ln;
124         this.first_Name = fn;
125     }
126
127     public String getInfo()
128     {
129         return "Name: " + this.first_Name + " " + this.last_Name;
130     }
131 }
132

```